

What is Networking?

Learning Objectives

- Define What is Networking? and explain its purpose
- Identify key components and features of What is Networking?
- Apply What is Networking? skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about What is Networking?? How is it used in real life?' Write student responses on the board. Guide discussion to connect What is Networking? to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of What is Networking?. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of What is Networking?. Define important terms and provide examples. Connect What is Networking? to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to What is Networking?. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of What is Networking? through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Networking? - mathematical concepts

Science: Scientific applications of What is Networking? - scientific applications

Language: Writing about What is Networking? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of What is Networking? concepts through practical exercises

Advanced: Apply What is Networking? concepts to solve complex problems independently

SEND: Practice What is Networking? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Networking? activity and oral questioning

CT Skill Assessed: Decomposition - breaking down What is Networking? into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Types of Networks

Learning Objectives

- Define Types of Networks and explain its purpose
- Identify key components and features of Types of Networks
- Apply Types of Networks skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Types of Networks? How is it used in real life?' Write student responses on the board. Guide discussion to connect Types of Networks to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Types of Networks. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Types of Networks. Define important terms and provide examples. Connect Types of Networks to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Types of Networks. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Types of Networks through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Types of Networks - mathematical concepts

Science: Scientific applications of Types of Networks - scientific applications

Language: Writing about Types of Networks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Types of Networks concepts through practical exercises

Advanced: Apply Types of Networks concepts to solve complex problems independently

SEND: Practice Types of Networks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Types of Networks activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Types of Networks into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Network Devices

Learning Objectives

- Define Network Devices and explain its purpose
- Identify key components and features of Network Devices
- Apply Network Devices skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Network Devices? How is it used in real life?' Write student responses on the board. Guide discussion to connect Network Devices to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Network Devices. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Network Devices. Define important terms and provide examples. Connect Network Devices to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Network Devices. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Network Devices through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Network Devices - mathematical concepts

Science: Scientific applications of Network Devices - scientific applications

Language: Writing about Network Devices builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Network Devices concepts through practical exercises

Advanced: Apply Network Devices concepts to solve complex problems independently

SEND: Practice Network Devices skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Network Devices activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Network Devices into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Internet Connection

Learning Objectives

- Define Internet Connection and explain its purpose
- Identify key components and features of Internet Connection
- Apply Internet Connection skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Internet Connection? How is it used in real life?' Write student responses on the board. Guide discussion to connect Internet Connection to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Internet Connection. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Internet Connection. Define important terms and provide examples. Connect Internet Connection to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Internet Connection. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Internet Connection through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Internet Connection - mathematical concepts

Science: Scientific applications of Internet Connection - scientific applications

Language: Writing about Internet Connection builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Internet Connection concepts through practical exercises

Advanced: Apply Internet Connection concepts to solve complex problems independently

SEND: Practice Internet Connection skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Internet Connection activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Internet Connection into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Internet Connection. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Internet Protocols

Learning Objectives

- Define Internet Protocols and explain its purpose
- Identify key components and features of Internet Protocols
- Apply Internet Protocols skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Internet Protocols? How is it used in real life?' Write student responses on the board. Guide discussion to connect Internet Protocols to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Internet Protocols. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Internet Protocols. Define important terms and provide examples. Connect Internet Protocols to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Internet Protocols. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Internet Protocols through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Internet Protocols - mathematical concepts

Science: Scientific applications of Internet Protocols - scientific applications

Language: Writing about Internet Protocols builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Internet Protocols concepts through practical exercises

Advanced: Apply Internet Protocols concepts to solve complex problems independently

SEND: Practice Internet Protocols skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Internet Protocols activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Internet Protocols into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Internet Protocols. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Email Services

Learning Objectives

- Define Email Services and explain its purpose
- Identify key components and features of Email Services
- Apply Email Services skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Email Services? How is it used in real life?' Write student responses on the board. Guide discussion to connect Email Services to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Email Services. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Email Services. Define important terms and provide examples. Connect Email Services to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Email Services. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Email Services through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Email Services - mathematical concepts

Science: Scientific applications of Email Services - scientific applications

Language: Writing about Email Services builds technical vocabulary and communication skills

Digital Citizen Reminder: Do not open emails from strangers - they may contain viruses.

Resources Needed:

- Research materials
- example AI applications
- discussion guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Email Services concepts through practical exercises

Advanced: Apply Email Services concepts to solve complex problems independently

SEND: Practice Email Services skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Email Services activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Email Services into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a paragraph explaining how Email Services is used in real life. Give two examples.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Attachments

Learning Objectives

- Define Attachments and explain its purpose
- Identify key components and features of Attachments
- Apply Attachments skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Attachments? How is it used in real life?' Write student responses on the board. Guide discussion to connect Attachments to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Attachments. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Attachments. Define important terms and provide examples. Connect Attachments to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Attachments. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Attachments through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: File sizes affect download time - understanding data transfer rates

Science: File formats serve different purposes - technology standards

Language: Writing clear attachment descriptions builds precise language

Digital Citizen Reminder: Only open attachments from people you trust - they may contain viruses.

Resources Needed:

- Email with attachments
- file type guide
- practice files

Differentiation

Approaching: Read an email and identify the attachment

On Level: Attach a file to an email and send it

Advanced: Attach multiple files of different formats with descriptions

SEND: Attach a file to an email with teacher guidance

Formative Assessment

Method: Attachment task

CT Skill Assessed: Algorithm Design - following steps to attach and send files

Dormitory Task (Paper-and-Pencil)

Write instructions for attaching a file to an email. List at least 5 steps.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Internet Safety

Learning Objectives

- Define Internet Safety and explain its purpose
- Identify key components and features of Internet Safety
- Apply Internet Safety skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Internet Safety? How is it used in real life?' Write student responses on the board. Guide discussion to connect Internet Safety to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Internet Safety. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Internet Safety. Define important terms and provide examples. Connect Internet Safety to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Internet Safety. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Internet Safety through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Risk assessment in math - probability and consequences

Science: Understanding online threats relates to biology (viruses as metaphor)

Language: Writing safety rules builds persuasive and instructional writing

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher. Use strong passwords with letters, numbers, and symbols.

Resources Needed:

- Safety rules poster
- scenario cards
- pledge template

Differentiation

Approaching: Match online safety rules to their correct category

On Level: Identify safe and unsafe behaviors in five online scenarios

Advanced: Create a personal internet safety plan with at least 8 rules

SEND: Discuss safety rules with teacher guidance using picture prompts

Formative Assessment

Method: Safety pledge and scenario quiz

CT Skill Assessed: Evaluation - assessing online situations for safety risks

Dormitory Task (Paper-and-Pencil)

Write ten internet safety rules. For each rule, explain what could happen if you do not follow it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Social Media

Learning Objectives

- Define Social Media and explain its purpose
- Identify key components and features of Social Media
- Apply Social Media skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Social Media? How is it used in real life?' Write student responses on the board. Guide discussion to connect Social Media to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Social Media. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Social Media. Define important terms and provide examples. Connect Social Media to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Social Media. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Social Media through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Social Media - mathematical concepts

Science: Scientific applications of Social Media - scientific applications

Language: Writing about Social Media builds technical vocabulary and communication skills

Digital Citizen Reminder: Be kind and respectful on social media.

Resources Needed:

- Chapter 1 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Social Media concepts through practical exercises

Advanced: Apply Social Media concepts to solve complex problems independently

SEND: Practice Social Media skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Social Media activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Social Media into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Social Media. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Download Management

Learning Objectives

- Define Download Management and explain its purpose
- Identify key components and features of Download Management
- Apply Download Management skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Download Management? How is it used in real life?' Write student responses on the board. Guide discussion to connect Download Management to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Download Management. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Download Management. Define important terms and provide examples. Connect Download Management to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Download Management. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Download Management through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Download Management - mathematical concepts

Science: Scientific applications of Download Management - scientific applications

Language: Writing about Download Management builds technical vocabulary and communication skills

Digital Citizen Reminder: Only download files from trusted websites.

Resources Needed:

- Chapter 1 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Download Management concepts through practical exercises

Advanced: Apply Download Management concepts to solve complex problems independently

SEND: Practice Download Management skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Download Management activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Download Management into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Download Management. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Network Troubleshooting

Learning Objectives

- Define Network Troubleshooting and explain its purpose
- Identify key components and features of Network Troubleshooting
- Apply Network Troubleshooting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Network Troubleshooting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Network Troubleshooting to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Network Troubleshooting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Network Troubleshooting. Define important terms and provide examples. Connect Network Troubleshooting to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Network Troubleshooting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Network Troubleshooting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Network Troubleshooting - mathematical concepts

Science: Scientific applications of Network Troubleshooting - scientific applications

Language: Writing about Network Troubleshooting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Network diagram
- connection equipment
- troubleshooting guide

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Network Troubleshooting concepts through practical exercises

Advanced: Apply Network Troubleshooting concepts to solve complex problems independently

SEND: Practice Network Troubleshooting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Network Troubleshooting activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Network Troubleshooting into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple network diagram showing how two computers connect through the internet.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 1 Review

Learning Objectives

- Define Chapter 1 Review and explain its purpose
- Identify key components and features of Chapter 1 Review
- Apply Chapter 1 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 1 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 1 Review to the chapter theme of Advanced Graphics.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 1 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 1 Review. Define important terms and provide examples. Connect Chapter 1 Review to the broader theme of Advanced Graphics. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 1 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 1 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises reinforce mathematical problem-solving

Science: Chapter review connects all internet and computing concepts

Language: Writing summaries builds synthesis and review skills

Digital Citizen Reminder: Apply internet safety rules you learned in every online activity.

Resources Needed:

- Review worksheet
- answer key
- chapter summary

Differentiation

Approaching: Complete fill-in-the-blank review questions

On Level: Answer short answer questions about chapter concepts

Advanced: Create a concept map connecting all chapter topics

SEND: Review key terms with teacher guidance using flashcards

Formative Assessment

Method: Chapter review quiz

CT Skill Assessed: Evaluation - self-assessing understanding of chapter concepts

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned in this chapter. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to GIMP

Learning Objectives

- Define Introduction to GIMP and explain its purpose
- Identify key components and features of Introduction to GIMP
- Apply Introduction to GIMP skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to GIMP? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to GIMP to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to GIMP. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to GIMP. Define important terms and provide examples. Connect Introduction to GIMP to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to GIMP. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to GIMP through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to GIMP - geometric transformations

Science: Scientific applications of Introduction to GIMP - digital imaging

Language: Writing about Introduction to GIMP builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Introduction to GIMP concepts through practical exercises

Advanced: Apply Introduction to GIMP concepts to solve complex problems independently

SEND: Practice Introduction to GIMP skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to GIMP activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Introduction to GIMP into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to GIMP. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Selection Tools

Learning Objectives

- Define Selection Tools and explain its purpose
- Identify key components and features of Selection Tools
- Apply Selection Tools skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Selection Tools? How is it used in real life?' Write student responses on the board. Guide discussion to connect Selection Tools to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Selection Tools. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Selection Tools. Define important terms and provide examples. Connect Selection Tools to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Selection Tools. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Selection Tools through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Selection Tools - geometric transformations

Science: Scientific applications of Selection Tools - digital imaging

Language: Writing about Selection Tools builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Selection Tools concepts through practical exercises

Advanced: Apply Selection Tools concepts to solve complex problems independently

SEND: Practice Selection Tools skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Selection Tools activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Selection Tools into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Selection Tools. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Layers Concept

Learning Objectives

- Define Layers Concept and explain its purpose
- Identify key components and features of Layers Concept
- Apply Layers Concept skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Layers Concept? How is it used in real life?' Write student responses on the board. Guide discussion to connect Layers Concept to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Layers Concept. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Layers Concept. Define important terms and provide examples. Connect Layers Concept to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Layers Concept. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Layers Concept through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Layers stack like fractions - understanding ordering and precedence

Science: How transparent layers work - optics and light

Language: Explaining layers requires clear organizational writing

Digital Citizen Reminder: Understanding layers helps organize work - a skill useful beyond computing.

Resources Needed:

- Computer with layer-capable editor
- layer examples
- practice files

Differentiation

Approaching: Create two layers and draw different elements on each

On Level: Use layers to create a composite image with foreground and background

Advanced: Manage complex multi-layer compositions with blending modes

SEND: Create and manage layers with teacher guidance

Formative Assessment

Method: Layer management demonstration

CT Skill Assessed: Abstraction - understanding that layers separate elements for independent editing

Dormitory Task (Paper-and-Pencil)

Draw three separate elements (sun, house, tree) on separate pieces of paper. Stack them to create one picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Layer Effects

Learning Objectives

- Define Layer Effects and explain its purpose
- Identify key components and features of Layer Effects
- Apply Layer Effects skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Layer Effects? How is it used in real life?' Write student responses on the board. Guide discussion to connect Layer Effects to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Layer Effects. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Layer Effects. Define important terms and provide examples. Connect Layer Effects to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Layer Effects. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Layer Effects through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Layer Effects - geometric transformations

Science: Scientific applications of Layer Effects - digital imaging

Language: Writing about Layer Effects builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Layer Effects concepts through practical exercises

Advanced: Apply Layer Effects concepts to solve complex problems independently

SEND: Practice Layer Effects skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Layer Effects activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Layer Effects into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Layer Effects. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Drawing Tools

Learning Objectives

- Define Drawing Tools and explain its purpose
- Identify key components and features of Drawing Tools
- Apply Drawing Tools skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Drawing Tools? How is it used in real life?' Write student responses on the board. Guide discussion to connect Drawing Tools to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Drawing Tools. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Drawing Tools. Define important terms and provide examples. Connect Drawing Tools to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Drawing Tools. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Drawing Tools through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Geometric shapes in art - identifying mathematical shapes in drawings

Science: Understanding brush strokes relates to fluid dynamics

Language: Describing drawing techniques builds technical vocabulary

Digital Citizen Reminder: Use drawing tools responsibly - do not draw on screens with markers or sharp objects.

Resources Needed:

- Computer with Paint
- tool practice sheets
- example artwork

Differentiation

Approaching: Use the line and rectangle tools to draw a building

On Level: Create a picture using at least 6 different drawing tools

Advanced: Combine advanced tools to create a detailed scene

SEND: Practice drawing with each tool and describe the results

Formative Assessment

Method: Drawing portfolio and tool checklist

CT Skill Assessed: Pattern Recognition - using repeated elements in designs

Dormitory Task (Paper-and-Pencil)

Draw a picture using only straight lines and rectangles. Count and write how many of each you used.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Color Adjustments

Learning Objectives

- Define Color Adjustments and explain its purpose
- Identify key components and features of Color Adjustments
- Apply Color Adjustments skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Color Adjustments? How is it used in real life?' Write student responses on the board. Guide discussion to connect Color Adjustments to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Color Adjustments. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Color Adjustments. Define important terms and provide examples. Connect Color Adjustments to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Color Adjustments. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Color Adjustments through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Color Adjustments - geometric transformations

Science: Scientific applications of Color Adjustments - digital imaging

Language: Writing about Color Adjustments builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Color Adjustments concepts through practical exercises

Advanced: Apply Color Adjustments concepts to solve complex problems independently

SEND: Practice Color Adjustments skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Color Adjustments activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Color Adjustments into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Color Adjustments. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Filters Introduction

Learning Objectives

- Define Filters Introduction and explain its purpose
- Identify key components and features of Filters Introduction
- Apply Filters Introduction skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Filters Introduction? How is it used in real life?' Write student responses on the board. Guide discussion to connect Filters Introduction to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Filters Introduction. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Filters Introduction. Define important terms and provide examples. Connect Filters Introduction to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Filters Introduction. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Filters Introduction through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Filters Introduction - geometric transformations

Science: Scientific applications of Filters Introduction - digital imaging

Language: Writing about Filters Introduction builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Filters Introduction concepts through practical exercises

Advanced: Apply Filters Introduction concepts to solve complex problems independently

SEND: Practice Filters Introduction skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Filters Introduction activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Filters Introduction into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Filters Introduction. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Artistic Filters

Learning Objectives

- Define Artistic Filters and explain its purpose
- Identify key components and features of Artistic Filters
- Apply Artistic Filters skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Artistic Filters? How is it used in real life?' Write student responses on the board. Guide discussion to connect Artistic Filters to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Artistic Filters. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Artistic Filters. Define important terms and provide examples. Connect Artistic Filters to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Artistic Filters. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Artistic Filters through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Artistic Filters - geometric transformations

Science: Scientific applications of Artistic Filters - digital imaging

Language: Writing about Artistic Filters builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Artistic Filters concepts through practical exercises

Advanced: Apply Artistic Filters concepts to solve complex problems independently

SEND: Practice Artistic Filters skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Artistic Filters activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Artistic Filters into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Artistic Filters. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Text and Typography

Learning Objectives

- Define Text and Typography and explain its purpose
- Identify key components and features of Text and Typography
- Apply Text and Typography skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Text and Typography? How is it used in real life?' Write student responses on the board. Guide discussion to connect Text and Typography to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Text and Typography. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Text and Typography. Define important terms and provide examples. Connect Text and Typography to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Text and Typography. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Text and Typography through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text and Typography - geometric transformations

Science: Scientific applications of Text and Typography - digital imaging

Language: Writing about Text and Typography builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Text and Typography concepts through practical exercises

Advanced: Apply Text and Typography concepts to solve complex problems independently

SEND: Practice Text and Typography skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text and Typography activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Text and Typography into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Text and Typography. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Image Composition

Learning Objectives

- Define Image Composition and explain its purpose
- Identify key components and features of Image Composition
- Apply Image Composition skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Image Composition? How is it used in real life?' Write student responses on the board. Guide discussion to connect Image Composition to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Image Composition. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Image Composition. Define important terms and provide examples. Connect Image Composition to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Image Composition. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Image Composition through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Image Composition - geometric transformations

Science: Scientific applications of Image Composition - digital imaging

Language: Writing about Image Composition builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Image Composition concepts through practical exercises

Advanced: Apply Image Composition concepts to solve complex problems independently

SEND: Practice Image Composition skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Image Composition activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Image Composition into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Photo Editing

Learning Objectives

- Define Project - Photo Editing and explain its purpose
- Identify key components and features of Project - Photo Editing
- Apply Project - Photo Editing skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Photo Editing? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Photo Editing to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Photo Editing. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Photo Editing. Define important terms and provide examples. Connect Project - Photo Editing to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Photo Editing. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Photo Editing through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Photo Editing - geometric transformations

Science: Scientific applications of Project - Photo Editing - digital imaging

Language: Writing about Project - Photo Editing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:

- Advanced Graphics reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support

On Level: Demonstrate understanding of Project - Photo Editing concepts through practical exercises

Advanced: Apply Project - Photo Editing concepts to solve complex problems independently

SEND: Practice Project - Photo Editing skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Photo Editing project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Photo Editing project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 2 Review

Learning Objectives

- Define Chapter 2 Review and explain its purpose
- Identify key components and features of Chapter 2 Review
- Apply Chapter 2 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 2 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 2 Review to the chapter theme of Word Processing Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 2 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 2 Review. Define important terms and provide examples. Connect Chapter 2 Review to the broader theme of Word Processing Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 2 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 2 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate programming concepts

Science: Chapter summaries connect related topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 2 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 2 Review support

On Level: Demonstrate Chapter 2 Review through practical exercises

Advanced: Apply Chapter 2 Review to solve complex problems independently

SEND: Practice Chapter 2 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 2 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 2 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 2 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

What is Multimedia?

Learning Objectives

- Define What is Multimedia? and explain its purpose
- Identify key components and features of What is Multimedia?
- Apply What is Multimedia? skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about What is Multimedia?? How is it used in real life?' Write student responses on the board. Guide discussion to connect What is Multimedia? to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of What is Multimedia?. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of What is Multimedia?. Define important terms and provide examples. Connect What is Multimedia? to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to What is Multimedia?. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of What is Multimedia? through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Multimedia? - network topology

Science: Scientific applications of What is Multimedia? - data transmission

Language: Writing about What is Multimedia? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of What is Multimedia? concepts through practical exercises

Advanced: Apply What is Multimedia? concepts to solve complex problems independently

SEND: Practice What is Multimedia? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Multimedia? activity and oral questioning

CT Skill Assessed: Decomposition - breaking down What is Multimedia? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about What is Multimedia?. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Audio Editing with Audacity

Learning Objectives

- Define Audio Editing with Audacity and explain its purpose
- Identify key components and features of Audio Editing with Audacity
- Apply Audio Editing with Audacity skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Audio Editing with Audacity? How is it used in real life?' Write student responses on the board. Guide discussion to connect Audio Editing with Audacity to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Audio Editing with Audacity. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Audio Editing with Audacity. Define important terms and provide examples. Connect Audio Editing with Audacity to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Audio Editing with Audacity. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Audio Editing with Audacity through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Editing with Audacity - network topology

Science: Scientific applications of Audio Editing with Audacity - data transmission

Language: Writing about Audio Editing with Audacity builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Audio Editing with Audacity concepts through practical exercises

Advanced: Apply Audio Editing with Audacity concepts to solve complex problems independently

SEND: Practice Audio Editing with Audacity skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Editing with Audacity activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Audio Editing with Audacity into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Audio Mixing

Learning Objectives

- Define Audio Mixing and explain its purpose
- Identify key components and features of Audio Mixing
- Apply Audio Mixing skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Audio Mixing? How is it used in real life?' Write student responses on the board. Guide discussion to connect Audio Mixing to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Audio Mixing. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Audio Mixing. Define important terms and provide examples. Connect Audio Mixing to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Audio Mixing. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Audio Mixing through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Mixing - network topology

Science: Scientific applications of Audio Mixing - data transmission

Language: Writing about Audio Mixing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Audio Mixing concepts through practical exercises

Advanced: Apply Audio Mixing concepts to solve complex problems independently

SEND: Practice Audio Mixing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Mixing activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Audio Mixing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Concepts

Learning Objectives

- Define Video Concepts and explain its purpose
- Identify key components and features of Video Concepts
- Apply Video Concepts skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Video Concepts? How is it used in real life?' Write student responses on the board. Guide discussion to connect Video Concepts to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Video Concepts. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Video Concepts. Define important terms and provide examples. Connect Video Concepts to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Video Concepts. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Video Concepts through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Concepts - network topology

Science: Scientific applications of Video Concepts - data transmission

Language: Writing about Video Concepts builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Video Concepts concepts through practical exercises

Advanced: Apply Video Concepts concepts to solve complex problems independently

SEND: Practice Video Concepts skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Concepts activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Video Concepts into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Video Editing Basics

Learning Objectives

- Define Video Editing Basics and explain its purpose
- Identify key components and features of Video Editing Basics
- Apply Video Editing Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Video Editing Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Video Editing Basics to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Video Editing Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Video Editing Basics. Define important terms and provide examples. Connect Video Editing Basics to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Video Editing Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Video Editing Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Editing Basics - network topology

Science: Scientific applications of Video Editing Basics - data transmission

Language: Writing about Video Editing Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Video Editing Basics concepts through practical exercises

Advanced: Apply Video Editing Basics concepts to solve complex problems independently

SEND: Practice Video Editing Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Editing Basics activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Video Editing Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Presentation Software

Learning Objectives

- Define Presentation Software and explain its purpose
- Identify key components and features of Presentation Software
- Apply Presentation Software skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Presentation Software? How is it used in real life?' Write student responses on the board. Guide discussion to connect Presentation Software to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Presentation Software. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Presentation Software. Define important terms and provide examples. Connect Presentation Software to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Presentation Software. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Presentation Software through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Presentation Software - network topology

Science: Scientific applications of Presentation Software - data transmission

Language: Writing about Presentation Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Presentation Software concepts through practical exercises

Advanced: Apply Presentation Software concepts to solve complex problems independently

SEND: Practice Presentation Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Presentation Software activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Presentation Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Presentation Software. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Slide Design

Learning Objectives

- Define Slide Design and explain its purpose
- Identify key components and features of Slide Design
- Apply Slide Design skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Slide Design? How is it used in real life?' Write student responses on the board. Guide discussion to connect Slide Design to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Slide Design. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Slide Design. Define important terms and provide examples. Connect Slide Design to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Slide Design. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Slide Design through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Slide Design - network topology

Science: Scientific applications of Slide Design - data transmission

Language: Writing about Slide Design builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Slide Design concepts through practical exercises

Advanced: Apply Slide Design concepts to solve complex problems independently

SEND: Practice Slide Design skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Slide Design activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Slide Design into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Slide Design. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Animations

Learning Objectives

- Define Animations and explain its purpose
- Identify key components and features of Animations
- Apply Animations skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Animations? How is it used in real life?' Write student responses on the board. Guide discussion to connect Animations to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Animations. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Animations. Define important terms and provide examples. Connect Animations to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Animations. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Animations through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Animations - network topology

Science: Scientific applications of Animations - data transmission

Language: Writing about Animations builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Animations concepts through practical exercises

Advanced: Apply Animations concepts to solve complex problems independently

SEND: Practice Animations skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Animations activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Animations into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Animations. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Slide Transitions

Learning Objectives

- Define Slide Transitions and explain its purpose
- Identify key components and features of Slide Transitions
- Apply Slide Transitions skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Slide Transitions? How is it used in real life?' Write student responses on the board. Guide discussion to connect Slide Transitions to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Slide Transitions. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Slide Transitions. Define important terms and provide examples. Connect Slide Transitions to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Slide Transitions. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Slide Transitions through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Slide Transitions - network topology

Science: Scientific applications of Slide Transitions - data transmission

Language: Writing about Slide Transitions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Slide Transitions concepts through practical exercises

Advanced: Apply Slide Transitions concepts to solve complex problems independently

SEND: Practice Slide Transitions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Slide Transitions activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Slide Transitions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Slide Transitions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Multimedia Integration

Learning Objectives

- Define Multimedia Integration and explain its purpose
- Identify key components and features of Multimedia Integration
- Apply Multimedia Integration skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Multimedia Integration? How is it used in real life?' Write student responses on the board. Guide discussion to connect Multimedia Integration to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Multimedia Integration. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Multimedia Integration. Define important terms and provide examples. Connect Multimedia Integration to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Multimedia Integration. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Multimedia Integration through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Multimedia Integration - network topology

Science: Scientific applications of Multimedia Integration - data transmission

Language: Writing about Multimedia Integration builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Multimedia Integration concepts through practical exercises

Advanced: Apply Multimedia Integration concepts to solve complex problems independently

SEND: Practice Multimedia Integration skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Multimedia Integration activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Multimedia Integration into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Multimedia Integration. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Presentation

Learning Objectives

- Define Project - Presentation and explain its purpose
- Identify key components and features of Project - Presentation
- Apply Project - Presentation skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Presentation? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Presentation to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Presentation. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Presentation. Define important terms and provide examples. Connect Project - Presentation to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Presentation. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Presentation through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Presentation - network topology

Science: Scientific applications of Project - Presentation - data transmission

Language: Writing about Project - Presentation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support

On Level: Demonstrate understanding of Project - Presentation concepts through practical exercises

Advanced: Apply Project - Presentation concepts to solve complex problems independently

SEND: Practice Project - Presentation skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Presentation project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Presentation project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 3 Review

Learning Objectives

- Define Chapter 3 Review and explain its purpose
- Identify key components and features of Chapter 3 Review
- Apply Chapter 3 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 3 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 3 Review to the chapter theme of Spreadsheets Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 3 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 3 Review. Define important terms and provide examples. Connect Chapter 3 Review to the broader theme of Spreadsheets Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 3 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 3 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate database concepts

Science: Chapter summaries connect SQL topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 3 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 3 Review support

On Level: Demonstrate Chapter 3 Review through practical exercises

Advanced: Apply Chapter 3 Review to solve complex problems independently

SEND: Practice Chapter 3 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 3 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 3 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 3 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Introduction to Spreadsheets

Learning Objectives

- Define Introduction to Spreadsheets and explain its purpose
- Identify key components and features of Introduction to Spreadsheets
- Apply Introduction to Spreadsheets skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Introduction to Spreadsheets? How is it used in real life?' Write student responses on the board. Guide discussion to connect Introduction to Spreadsheets to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Introduction to Spreadsheets. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Introduction to Spreadsheets. Define important terms and provide examples. Connect Introduction to Spreadsheets to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Introduction to Spreadsheets. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Introduction to Spreadsheets through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Cells use row-column addresses like coordinates - math connection

Science: Spreadsheets organize data like scientific tables

Language: Writing clear data labels builds precise language

Digital Citizen Reminder: Do not enter personal information into shared spreadsheets.

Resources Needed:

- Computer with spreadsheet software
- sample data
- practice worksheet

Differentiation

Approaching: Enter text and numbers into cells

On Level: Create a simple data table with headers and data

Advanced: Design a complex spreadsheet with multiple data types and formatting

SEND: Enter data into cells with teacher guidance

Formative Assessment

Method: Data entry accuracy check

CT Skill Assessed: Decomposition - understanding rows, columns, and cells

Dormitory Task (Paper-and-Pencil)

Draw a grid with 5 columns and 6 rows on graph paper. Label the columns and fill in data about your class.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Basic Formulas

Learning Objectives

- Define Basic Formulas and explain its purpose
- Identify key components and features of Basic Formulas
- Apply Basic Formulas skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Basic Formulas? How is it used in real life?' Write student responses on the board. Guide discussion to connect Basic Formulas to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Basic Formulas. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Basic Formulas. Define important terms and provide examples. Connect Basic Formulas to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Basic Formulas. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Basic Formulas through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Basic Formulas - mathematical concepts

Science: Scientific applications of Basic Formulas - scientific applications

Language: Writing about Basic Formulas builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Basic Formulas concepts through practical exercises

Advanced: Apply Basic Formulas concepts to solve complex problems independently

SEND: Practice Basic Formulas skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Basic Formulas activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Basic Formulas into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Basic Formulas. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Cell References

Learning Objectives

- Define Cell References and explain its purpose
- Identify key components and features of Cell References
- Apply Cell References skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Cell References? How is it used in real life?' Write student responses on the board. Guide discussion to connect Cell References to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Cell References. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Cell References. Define important terms and provide examples. Connect Cell References to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Cell References. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Cell References through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Cell References - mathematical concepts

Science: Scientific applications of Cell References - scientific applications

Language: Writing about Cell References builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Cell References concepts through practical exercises

Advanced: Apply Cell References concepts to solve complex problems independently

SEND: Practice Cell References skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Cell References activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Cell References into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Cell References. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Functions

Learning Objectives

- Define Functions and explain its purpose
- Identify key components and features of Functions
- Apply Functions skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Functions? How is it used in real life?' Write student responses on the board. Guide discussion to connect Functions to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Functions. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Functions. Define important terms and provide examples. Connect Functions to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Functions. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Functions through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Functions - mathematical concepts

Science: Scientific applications of Functions - scientific applications

Language: Writing about Functions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Functions concepts through practical exercises

Advanced: Apply Functions concepts to solve complex problems independently

SEND: Practice Functions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Functions activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Functions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Functions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Charts Creation

Learning Objectives

- Define Charts Creation and explain its purpose
- Identify key components and features of Charts Creation
- Apply Charts Creation skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Charts Creation? How is it used in real life?' Write student responses on the board. Guide discussion to connect Charts Creation to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Charts Creation. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Charts Creation. Define important terms and provide examples. Connect Charts Creation to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Charts Creation. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Charts Creation through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Charts Creation - mathematical concepts

Science: Scientific applications of Charts Creation - scientific applications

Language: Writing about Charts Creation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Charts Creation concepts through practical exercises

Advanced: Apply Charts Creation concepts to solve complex problems independently

SEND: Practice Charts Creation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Charts Creation activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Charts Creation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Charts Creation. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chart Formatting

Learning Objectives

- Define Chart Formatting and explain its purpose
- Identify key components and features of Chart Formatting
- Apply Chart Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chart Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chart Formatting to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chart Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chart Formatting. Define important terms and provide examples. Connect Chart Formatting to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chart Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chart Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Chart Formatting - mathematical concepts

Science: Scientific applications of Chart Formatting - scientific applications

Language: Writing about Chart Formatting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Chart Formatting concepts through practical exercises

Advanced: Apply Chart Formatting concepts to solve complex problems independently

SEND: Practice Chart Formatting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Chart Formatting activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Chart Formatting into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Chart Formatting. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Sorting

Learning Objectives

- Define Data Sorting and explain its purpose
- Identify key components and features of Data Sorting
- Apply Data Sorting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Sorting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Sorting to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Sorting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Sorting. Define important terms and provide examples. Connect Data Sorting to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Sorting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Sorting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Sorting - mathematical concepts

Science: Scientific applications of Data Sorting - scientific applications

Language: Writing about Data Sorting builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Data Sorting concepts through practical exercises

Advanced: Apply Data Sorting concepts to solve complex problems independently

SEND: Practice Data Sorting skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Sorting activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Data Sorting into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Sorting. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Filtering

Learning Objectives

- Define Data Filtering and explain its purpose
- Identify key components and features of Data Filtering
- Apply Data Filtering skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Filtering? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Filtering to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Filtering. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Filtering. Define important terms and provide examples. Connect Data Filtering to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Filtering. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Filtering through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Filtering - mathematical concepts

Science: Scientific applications of Data Filtering - scientific applications

Language: Writing about Data Filtering builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Data Filtering concepts through practical exercises

Advanced: Apply Data Filtering concepts to solve complex problems independently

SEND: Practice Data Filtering skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Filtering activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Data Filtering into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Filtering. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Data Validation

Learning Objectives

- Define Data Validation and explain its purpose
- Identify key components and features of Data Validation
- Apply Data Validation skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Data Validation? How is it used in real life?' Write student responses on the board. Guide discussion to connect Data Validation to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Data Validation. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Data Validation. Define important terms and provide examples. Connect Data Validation to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Data Validation. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Data Validation through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Data validation uses rules and constraints - logical conditions

Science: Error prevention through validation - quality control

Language: Writing validation rules builds precise specification skills

Digital Citizen Reminder: Data validation prevents errors - always set up rules for data entry.

Resources Needed:

- Spreadsheet
- validation examples
- practice exercises

Differentiation

Approaching: Add a dropdown list to a cell

On Level: Create data validation with custom error messages

Advanced: Design comprehensive validation rules for a complete data entry system

SEND: Add a dropdown list with teacher guidance

Formative Assessment

Method: Data validation setup task

CT Skill Assessed: Decomposition - breaking validation rules into specific conditions

Dormitory Task (Paper-and-Pencil)

Write three data validation rules for a student marks spreadsheet. Example: Marks must be between 0 and 100.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Conditional Formatting

Learning Objectives

- Define Conditional Formatting and explain its purpose
- Identify key components and features of Conditional Formatting
- Apply Conditional Formatting skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Conditional Formatting? How is it used in real life?' Write student responses on the board. Guide discussion to connect Conditional Formatting to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Conditional Formatting. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Conditional Formatting. Define important terms and provide examples. Connect Conditional Formatting to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Conditional Formatting. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Conditional Formatting through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: IF conditions use logical operators (>, <, =) - Boolean math

Science: Conditional formatting highlights patterns in data

Language: Writing IF conditions builds logical thinking skills

Digital Citizen Reminder: Use conditional formatting to highlight important data, not to hide it.

Resources Needed:

- Spreadsheet with data
- condition examples
- practice exercises

Differentiation

Approaching: Apply color to cells that meet one condition

On Level: Use IF function to display different text based on values

Advanced: Apply multiple conditional formatting rules with complex conditions

SEND: Apply a simple condition with teacher guidance

Formative Assessment

Method: Conditional formatting task

CT Skill Assessed: Pattern Recognition - identifying data patterns through visual highlighting

Dormitory Task (Paper-and-Pencil)

Write 5 conditions and their outcomes. Example: If score > 90, display Excellent.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Grade Sheet

Learning Objectives

- Define Project - Grade Sheet and explain its purpose
- Identify key components and features of Project - Grade Sheet
- Apply Project - Grade Sheet skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Grade Sheet? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Grade Sheet to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Grade Sheet. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Grade Sheet. Define important terms and provide examples. Connect Project - Grade Sheet to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Grade Sheet. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Grade Sheet through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Grade Sheet - mathematical concepts

Science: Scientific applications of Project - Grade Sheet - scientific applications

Language: Writing about Project - Grade Sheet builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Grade Sheet concepts through practical exercises

Advanced: Apply Project - Grade Sheet concepts to solve complex problems independently

SEND: Practice Project - Grade Sheet skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Grade Sheet project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Grade Sheet project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 4 Review

Learning Objectives

- Define Chapter 4 Review and explain its purpose
- Identify key components and features of Chapter 4 Review
- Apply Chapter 4 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 4 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 4 Review to the chapter theme of Presentations Advanced.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 4 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 4 Review. Define important terms and provide examples. Connect Chapter 4 Review to the broader theme of Presentations Advanced. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 4 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 4 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate web development concepts

Science: Chapter summaries connect HTML and CSS topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 4 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 4 Review support

On Level: Demonstrate Chapter 4 Review through practical exercises

Advanced: Apply Chapter 4 Review to solve complex problems independently

SEND: Practice Chapter 4 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 4 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 4 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 4 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Scratch Review

Learning Objectives

- Define Scratch Review and explain its purpose
- Identify key components and features of Scratch Review
- Apply Scratch Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Scratch Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Scratch Review to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Scratch Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Scratch Review. Define important terms and provide examples. Connect Scratch Review to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Scratch Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Scratch Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Scratch Review - mathematical concepts

Science: Scientific applications of Scratch Review - scientific applications

Language: Writing about Scratch Review builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Scratch Review concepts through practical exercises

Advanced: Apply Scratch Review concepts to solve complex problems independently

SEND: Practice Scratch Review skills with step-by-step teacher guidance

Formative Assessment

Method: Chapter review quiz and self-assessment

CT Skill Assessed: Evaluation - self-assessment of Chapter 5 learning

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned about Chapter 5. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Variables Advanced

Learning Objectives

- Define Variables Advanced and explain its purpose
- Identify key components and features of Variables Advanced
- Apply Variables Advanced skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Variables Advanced? How is it used in real life?' Write student responses on the board. Guide discussion to connect Variables Advanced to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Variables Advanced. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Variables Advanced. Define important terms and provide examples. Connect Variables Advanced to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Variables Advanced. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Variables Advanced through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Variables Advanced - mathematical concepts

Science: Scientific applications of Variables Advanced - scientific applications

Language: Writing about Variables Advanced builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Variables Advanced concepts through practical exercises

Advanced: Apply Variables Advanced concepts to solve complex problems independently

SEND: Practice Variables Advanced skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Variables Advanced activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Variables Advanced into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Variables Advanced. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Conditional Logic

Learning Objectives

- Define Conditional Logic and explain its purpose
- Identify key components and features of Conditional Logic
- Apply Conditional Logic skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Conditional Logic? How is it used in real life?' Write student responses on the board. Guide discussion to connect Conditional Logic to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Conditional Logic. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Conditional Logic. Define important terms and provide examples. Connect Conditional Logic to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Conditional Logic. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Conditional Logic through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Conditional Logic - mathematical concepts

Science: Scientific applications of Conditional Logic - scientific applications

Language: Writing about Conditional Logic builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Conditional Logic concepts through practical exercises

Advanced: Apply Conditional Logic concepts to solve complex problems independently

SEND: Practice Conditional Logic skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Conditional Logic activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Conditional Logic into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Conditional Logic. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Loops Advanced

Learning Objectives

- Define Loops Advanced and explain its purpose
- Identify key components and features of Loops Advanced
- Apply Loops Advanced skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Loops Advanced? How is it used in real life?' Write student responses on the board. Guide discussion to connect Loops Advanced to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Loops Advanced. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Loops Advanced. Define important terms and provide examples. Connect Loops Advanced to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Loops Advanced. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Loops Advanced through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Loops Advanced - mathematical concepts

Science: Scientific applications of Loops Advanced - scientific applications

Language: Writing about Loops Advanced builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Loops Advanced concepts through practical exercises

Advanced: Apply Loops Advanced concepts to solve complex problems independently

SEND: Practice Loops Advanced skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Loops Advanced activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Loops Advanced into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Loops Advanced. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Sensing Blocks

Learning Objectives

- Define Sensing Blocks and explain its purpose
- Identify key components and features of Sensing Blocks
- Apply Sensing Blocks skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Sensing Blocks? How is it used in real life?' Write student responses on the board. Guide discussion to connect Sensing Blocks to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Sensing Blocks. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Sensing Blocks. Define important terms and provide examples. Connect Sensing Blocks to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Sensing Blocks. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Sensing Blocks through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sensing Blocks - mathematical concepts

Science: Scientific applications of Sensing Blocks - scientific applications

Language: Writing about Sensing Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Sensing Blocks concepts through practical exercises

Advanced: Apply Sensing Blocks concepts to solve complex problems independently

SEND: Practice Sensing Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sensing Blocks activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Sensing Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sensing Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Sprite Interactions

Learning Objectives

- Define Sprite Interactions and explain its purpose
- Identify key components and features of Sprite Interactions
- Apply Sprite Interactions skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Sprite Interactions? How is it used in real life?' Write student responses on the board. Guide discussion to connect Sprite Interactions to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Sprite Interactions. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Sprite Interactions. Define important terms and provide examples. Connect Sprite Interactions to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Sprite Interactions. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Sprite Interactions through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sprite Interactions - mathematical concepts

Science: Scientific applications of Sprite Interactions - scientific applications

Language: Writing about Sprite Interactions builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Sprite Interactions concepts through practical exercises

Advanced: Apply Sprite Interactions concepts to solve complex problems independently

SEND: Practice Sprite Interactions skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sprite Interactions activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Sprite Interactions into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sprite Interactions. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Clones

Learning Objectives

- Define Clones and explain its purpose
- Identify key components and features of Clones
- Apply Clones skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Clones? How is it used in real life?' Write student responses on the board. Guide discussion to connect Clones to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Clones. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Clones. Define important terms and provide examples. Connect Clones to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Clones. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Clones through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Clones - mathematical concepts

Science: Scientific applications of Clones - scientific applications

Language: Writing about Clones builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Clones concepts through practical exercises

Advanced: Apply Clones concepts to solve complex problems independently

SEND: Practice Clones skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Clones activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Clones into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Clones. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Game Design

Learning Objectives

- Define Game Design and explain its purpose
- Identify key components and features of Game Design
- Apply Game Design skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Game Design? How is it used in real life?' Write student responses on the board. Guide discussion to connect Game Design to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Game Design. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Game Design. Define important terms and provide examples. Connect Game Design to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Game Design. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Game Design through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Game Design - mathematical concepts

Science: Scientific applications of Game Design - scientific applications

Language: Writing about Game Design builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Game Design concepts through practical exercises

Advanced: Apply Game Design concepts to solve complex problems independently

SEND: Practice Game Design skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Game Design activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Game Design into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Game Design. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Keyboard Controls

Learning Objectives

- Define Keyboard Controls and explain its purpose
- Identify key components and features of Keyboard Controls
- Apply Keyboard Controls skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Keyboard Controls? How is it used in real life?' Write student responses on the board. Guide discussion to connect Keyboard Controls to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Keyboard Controls. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Keyboard Controls. Define important terms and provide examples. Connect Keyboard Controls to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Keyboard Controls. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Keyboard Controls through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Keyboard Controls - mathematical concepts

Science: Scientific applications of Keyboard Controls - scientific applications

Language: Writing about Keyboard Controls builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Keyboard Controls concepts through practical exercises

Advanced: Apply Keyboard Controls concepts to solve complex problems independently

SEND: Practice Keyboard Controls skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Keyboard Controls activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Keyboard Controls into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Keyboard Controls. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Random Events

Learning Objectives

- Define Random Events and explain its purpose
- Identify key components and features of Random Events
- Apply Random Events skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Random Events? How is it used in real life?' Write student responses on the board. Guide discussion to connect Random Events to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Random Events. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Random Events. Define important terms and provide examples. Connect Random Events to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Random Events. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Random Events through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Random Events - mathematical concepts

Science: Scientific applications of Random Events - scientific applications

Language: Writing about Random Events builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Random Events concepts through practical exercises

Advanced: Apply Random Events concepts to solve complex problems independently

SEND: Practice Random Events skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Random Events activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Random Events into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Random Events. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Game

Learning Objectives

- Define Project - Game and explain its purpose
- Identify key components and features of Project - Game
- Apply Project - Game skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Game? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Game to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Game. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Game. Define important terms and provide examples. Connect Project - Game to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Game. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Game through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Game - mathematical concepts

Science: Scientific applications of Project - Game - scientific applications

Language: Writing about Project - Game builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Game concepts through practical exercises

Advanced: Apply Project - Game concepts to solve complex problems independently

SEND: Practice Project - Game skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Game project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Game project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 5 Review

Learning Objectives

- Define Chapter 5 Review and explain its purpose
- Identify key components and features of Chapter 5 Review
- Apply Chapter 5 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 5 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 5 Review to the chapter theme of Database Introduction.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 5 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 5 Review. Define important terms and provide examples. Connect Chapter 5 Review to the broader theme of Database Introduction. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 5 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 5 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate algorithm concepts

Science: Chapter summaries connect all algorithm topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 5 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 5 Review support

On Level: Demonstrate Chapter 5 Review through practical exercises

Advanced: Apply Chapter 5 Review to solve complex problems independently

SEND: Practice Chapter 5 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 5 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 5 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 5 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

What is Digital Citizenship?

Learning Objectives

- Define What is Digital Citizenship? and explain its purpose
- Identify key components and features of What is Digital Citizenship?
- Apply What is Digital Citizenship? skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about What is Digital Citizenship?? How is it used in real life?' Write student responses on the board. Guide discussion to connect What is Digital Citizenship? to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of What is Digital Citizenship?. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of What is Digital Citizenship?. Define important terms and provide examples. Connect What is Digital Citizenship? to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to What is Digital Citizenship?. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of What is Digital Citizenship? through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Digital Citizenship? - mathematical concepts

Science: Scientific applications of What is Digital Citizenship? - scientific applications

Language: Writing about What is Digital Citizenship? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills responsibly and ethically.

Resources Needed:

- Scenario cards
- discussion guide
- pledge template

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of What is Digital Citizenship? concepts through practical exercises

Advanced: Apply What is Digital Citizenship? concepts to solve complex problems independently

SEND: Practice What is Digital Citizenship? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Digital Citizenship? activity and oral questioning

CT Skill Assessed: Decomposition - breaking down What is Digital Citizenship? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 rules for being a responsible digital citizen. Explain why each rule is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Online Safety Rules

Learning Objectives

- Define Online Safety Rules and explain its purpose
- Identify key components and features of Online Safety Rules
- Apply Online Safety Rules skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Online Safety Rules? How is it used in real life?' Write student responses on the board. Guide discussion to connect Online Safety Rules to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Online Safety Rules. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Online Safety Rules. Define important terms and provide examples. Connect Online Safety Rules to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Online Safety Rules. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Online Safety Rules through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Online Safety Rules - mathematical concepts

Science: Scientific applications of Online Safety Rules - scientific applications

Language: Writing about Online Safety Rules builds technical vocabulary and communication skills

Digital Citizen Reminder: Follow all online safety rules to protect yourself and your information.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Online Safety Rules concepts through practical exercises

Advanced: Apply Online Safety Rules concepts to solve complex problems independently

SEND: Practice Online Safety Rules skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Online Safety Rules activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Online Safety Rules into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Online Safety Rules. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Password Security

Learning Objectives

- Define Password Security and explain its purpose
- Identify key components and features of Password Security
- Apply Password Security skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Password Security? How is it used in real life?' Write student responses on the board. Guide discussion to connect Password Security to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Password Security. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Password Security. Define important terms and provide examples. Connect Password Security to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Password Security. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Password Security through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Password Security - mathematical concepts

Science: Scientific applications of Password Security - scientific applications

Language: Writing about Password Security builds technical vocabulary and communication skills

Digital Citizen Reminder: Use strong passwords with at least 8 characters including letters, numbers, and symbols.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Password Security concepts through practical exercises

Advanced: Apply Password Security concepts to solve complex problems independently

SEND: Practice Password Security skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Password Security activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Password Security into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Password Security. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Cyberbullying

Learning Objectives

- Define Cyberbullying and explain its purpose
- Identify key components and features of Cyberbullying
- Apply Cyberbullying skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Cyberbullying? How is it used in real life?' Write student responses on the board. Guide discussion to connect Cyberbullying to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Cyberbullying. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Cyberbullying. Define important terms and provide examples. Connect Cyberbullying to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Cyberbullying. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Cyberbullying through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Cyberbullying - mathematical concepts

Science: Scientific applications of Cyberbullying - scientific applications

Language: Writing about Cyberbullying builds technical vocabulary and communication skills

Digital Citizen Reminder: Never cyberbully others - treat everyone with respect online.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Cyberbullying concepts through practical exercises

Advanced: Apply Cyberbullying concepts to solve complex problems independently

SEND: Practice Cyberbullying skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Cyberbullying activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Cyberbullying into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Cyberbullying. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Digital Footprint

Learning Objectives

- Define Digital Footprint and explain its purpose
- Identify key components and features of Digital Footprint
- Apply Digital Footprint skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Digital Footprint? How is it used in real life?' Write student responses on the board. Guide discussion to connect Digital Footprint to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Digital Footprint. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Digital Footprint. Define important terms and provide examples. Connect Digital Footprint to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Digital Footprint. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Digital Footprint through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Digital Footprint - mathematical concepts

Science: Scientific applications of Digital Footprint - scientific applications

Language: Writing about Digital Footprint builds technical vocabulary and communication skills

Digital Citizen Reminder: Your digital footprint lasts forever - think before you post.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Digital Footprint concepts through practical exercises

Advanced: Apply Digital Footprint concepts to solve complex problems independently

SEND: Practice Digital Footprint skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Digital Footprint activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Digital Footprint into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Digital Footprint. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Online Predators

Learning Objectives

- Define Online Predators and explain its purpose
- Identify key components and features of Online Predators
- Apply Online Predators skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Online Predators? How is it used in real life?' Write student responses on the board. Guide discussion to connect Online Predators to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Online Predators. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Online Predators. Define important terms and provide examples. Connect Online Predators to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Online Predators. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Online Predators through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Online Predators - mathematical concepts

Science: Scientific applications of Online Predators - scientific applications

Language: Writing about Online Predators builds technical vocabulary and communication skills

Digital Citizen Reminder: Never meet online strangers in person - always tell a parent or teacher.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Online Predators concepts through practical exercises

Advanced: Apply Online Predators concepts to solve complex problems independently

SEND: Practice Online Predators skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Online Predators activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Online Predators into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Online Predators. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Copyright Basics

Learning Objectives

- Define Copyright Basics and explain its purpose
- Identify key components and features of Copyright Basics
- Apply Copyright Basics skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Copyright Basics? How is it used in real life?' Write student responses on the board. Guide discussion to connect Copyright Basics to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Copyright Basics. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Copyright Basics. Define important terms and provide examples. Connect Copyright Basics to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Copyright Basics. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Copyright Basics through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Copyright Basics - mathematical concepts

Science: Scientific applications of Copyright Basics - scientific applications

Language: Writing about Copyright Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Respect copyright - only use content you have permission to use.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Copyright Basics concepts through practical exercises

Advanced: Apply Copyright Basics concepts to solve complex problems independently

SEND: Practice Copyright Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Copyright Basics activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Copyright Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Copyright Basics. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Plagiarism

Learning Objectives

- Define Plagiarism and explain its purpose
- Identify key components and features of Plagiarism
- Apply Plagiarism skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Plagiarism? How is it used in real life?' Write student responses on the board. Guide discussion to connect Plagiarism to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Plagiarism. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Plagiarism. Define important terms and provide examples. Connect Plagiarism to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Plagiarism. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Plagiarism through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Plagiarism - mathematical concepts

Science: Scientific applications of Plagiarism - scientific applications

Language: Writing about Plagiarism builds technical vocabulary and communication skills

Digital Citizen Reminder: Always give credit to original authors - plagiarism is wrong.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Plagiarism concepts through practical exercises

Advanced: Apply Plagiarism concepts to solve complex problems independently

SEND: Practice Plagiarism skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Plagiarism activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Plagiarism into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Plagiarism. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Phishing Scams

Learning Objectives

- Define Phishing Scams and explain its purpose
- Identify key components and features of Phishing Scams
- Apply Phishing Scams skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Phishing Scams? How is it used in real life?' Write student responses on the board. Guide discussion to connect Phishing Scams to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Phishing Scams. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Phishing Scams. Define important terms and provide examples. Connect Phishing Scams to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Phishing Scams. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Phishing Scams through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Phishing Scams - mathematical concepts

Science: Scientific applications of Phishing Scams - scientific applications

Language: Writing about Phishing Scams builds technical vocabulary and communication skills

Digital Citizen Reminder: Be suspicious of messages asking for personal information - they may be scams.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Phishing Scams concepts through practical exercises

Advanced: Apply Phishing Scams concepts to solve complex problems independently

SEND: Practice Phishing Scams skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Phishing Scams activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Phishing Scams into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Phishing Scams. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Screen Time Management

Learning Objectives

- Define Screen Time Management and explain its purpose
- Identify key components and features of Screen Time Management
- Apply Screen Time Management skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Screen Time Management? How is it used in real life?' Write student responses on the board. Guide discussion to connect Screen Time Management to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Screen Time Management. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Screen Time Management. Define important terms and provide examples. Connect Screen Time Management to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Screen Time Management. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Screen Time Management through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Screen Time Management - mathematical concepts

Science: Scientific applications of Screen Time Management - scientific applications

Language: Writing about Screen Time Management builds technical vocabulary and communication skills

Digital Citizen Reminder: Balance screen time with outdoor activities for healthy development.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Screen Time Management concepts through practical exercises

Advanced: Apply Screen Time Management concepts to solve complex problems independently

SEND: Practice Screen Time Management skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Screen Time Management activity and oral questioning

CT Skill Assessed: Decomposition - breaking down Screen Time Management into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Screen Time Management. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Project - Safety Poster

Learning Objectives

- Define Project - Safety Poster and explain its purpose
- Identify key components and features of Project - Safety Poster
- Apply Project - Safety Poster skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Project - Safety Poster? How is it used in real life?' Write student responses on the board. Guide discussion to connect Project - Safety Poster to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Project - Safety Poster. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Project - Safety Poster. Define important terms and provide examples. Connect Project - Safety Poster to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Project - Safety Poster. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Project - Safety Poster through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Safety Poster - mathematical concepts

Science: Scientific applications of Project - Safety Poster - scientific applications

Language: Writing about Project - Safety Poster builds technical vocabulary and communication skills

Digital Citizen Reminder: Follow all online safety rules to protect yourself and your information.

Resources Needed:

- Chapter 6 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support

On Level: Demonstrate understanding of Project - Safety Poster concepts through practical exercises

Advanced: Apply Project - Safety Poster concepts to solve complex problems independently

SEND: Practice Project - Safety Poster skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment

CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Safety Poster project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Safety Poster project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Chapter 6 Review

Learning Objectives

- Define Chapter 6 Review and explain its purpose
- Identify key components and features of Chapter 6 Review
- Apply Chapter 6 Review skills in a practical task

Engage (5-7 minutes)

Teacher Activity

Ask: 'What do you already know about Chapter 6 Review? How is it used in real life?' Write student responses on the board. Guide discussion to connect Chapter 6 Review to the chapter theme of Scratch Programming.

Student Activity

Share prior knowledge about the topic. Listen to classmates ideas. Identify connections to daily life.

Explore (10-12 minutes)

Teacher Activity

Show examples and demonstrations of Chapter 6 Review. Guide students through hands-on exploration. Ask questions to deepen understanding: 'What do you notice? How does this work? Why is this important?'

Student Activity

Observe the demonstrations. Take notes on key observations. Discuss findings with a partner.

Explain (10-12 minutes)

Teacher Activity

Explain the key concepts of Chapter 6 Review. Define important terms and provide examples. Connect Chapter 6 Review to the broader theme of Scratch Programming. Use diagrams and real-world examples to clarify.

Student Activity

Take notes on key concepts and definitions. Draw diagrams. Write examples of how {topic} is applied.

Elaborate (8-10 minutes)

Teacher Activity

Present a challenge or project related to Chapter 6 Review. Students apply what they learned in a practical task. Encourage creativity and problem-solving.

Student Activity

Work on the challenge independently or in pairs. Apply concepts to create something new. Share results with the class.

Evaluate (5-8 minutes)

Teacher Activity

Assess understanding of Chapter 6 Review through an exit ticket or practical demonstration. Ask students to explain key concepts in their own words.

Student Activity

Complete the assessment. Explain what you learned about {topic}. Reflect on areas for improvement.

Cross-Curricular Connections

Mathematics: Review exercises consolidate technology concepts

Science: Chapter summaries connect all emerging tech topics

Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 6 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 6 Review support

On Level: Demonstrate Chapter 6 Review through practical exercises

Advanced: Apply Chapter 6 Review to solve complex problems independently

SEND: Practice Chapter 6 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 6 Review activity

CT Skill Assessed: Decomposition - breaking down Chapter 6 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 6 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: